

8

BIFURCATIONS REVISITED

8.0 Introduction

This chapter extends our earlier work on bifurcations (Chapter 3). As we move up from one-dimensional to two-dimensional systems, we still find that fixed points can be created or destroyed or destabilized as parameters are varied—but now the same is true of closed orbits as well. Thus we can begin to *describe the ways in which oscillations can be turned on or off*.

In this broader context, what exactly do we mean by a bifurcation? The usual definition involves the concept of “topological equivalence” (Section 6.3): if the phase portrait changes its topological structure as a parameter is varied, we say that a **bifurcation** has occurred. Examples include changes in the number or stability of fixed points, closed orbits, or saddle connections as a parameter is varied.

This chapter is organized as follows: for each bifurcation, we start with a simple prototypical example, and then graduate to more challenging examples, either briefly or in separate sections. Models of genetic switches, chemical oscillators, driven pendula and Josephson junctions are used to illustrate the theory.

8.1 Saddle-Node, Transcritical, and Pitchfork Bifurcations

The bifurcations of fixed points discussed in Chapter 3 have analogs in two dimensions (and indeed, in *all* dimensions). Yet it turns out that nothing really new happens when more dimensions are added—all the action is confined to a one-dimensional subspace along which the bifurcations occur, while in the extra dimensions the flow is either simple attraction or repulsion from that subspace, as we’ll see below.

Saddle-Node Bifurcation

The saddle-node bifurcation is the basic mechanism for the creation and destruction of fixed points. Here's the prototypical example in two dimensions:

$$\begin{aligned}\dot{x} &= \mu - x^2 \\ \dot{y} &= -y.\end{aligned}\tag{1}$$

In the x -direction we see the bifurcation behavior discussed in Section 3.1, while in the y -direction the motion is exponentially damped.

Consider the phase portrait as μ varies. For $\mu > 0$, Figure 8.1.1 shows that there are two fixed points, a stable node at $(x^*, y^*) = (\sqrt{\mu}, 0)$ and a saddle at $(-\sqrt{\mu}, 0)$. As μ decreases, the saddle and node approach each other, then collide when $\mu = 0$, and finally disappear when $\mu < 0$.

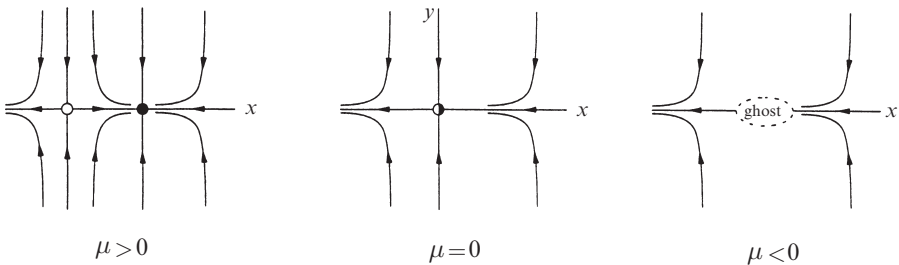


Figure 8.1.1

Even after the fixed points have annihilated each other, they continue to influence the flow—as in Section 4.3, they leave a *ghost*, a bottleneck region that sucks trajectories in and delays them before allowing passage out the other side. For the same reasons as in Section 4.3, the time spent in the bottleneck generically increases as $(\mu - \mu_c)^{-1/2}$, where μ_c is the value at which the saddle-node bifurcation occurs. Some applications of this scaling law in condensed-matter physics are

discussed by Strogatz and Westervelt (1989).

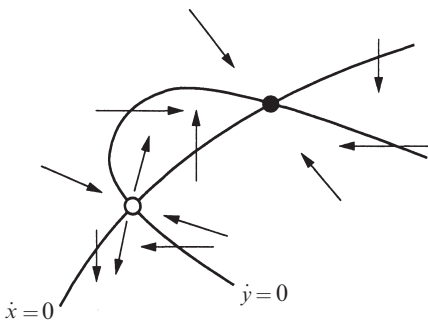


Figure 8.1.2

Figure 8.1.1 is representative of the following more general situation. Consider a two-dimensional system $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$ that depends on a parameter μ . Suppose that for some value of μ the nullclines intersect as shown in Figure 8.1.2. Notice that each intersection corresponds to a fixed point since $\dot{x} = 0$ and $\dot{y} = 0$ simultane-

ously. Thus, to see how the fixed points move as μ changes, we just have to watch the intersections. Now suppose that the nullclines pull away from each other as μ varies, becoming *tangent* at $\mu = \mu_c$. Then the fixed points approach each other and collide when $\mu = \mu_c$; after the nullclines pull apart, there are no intersections and the fixed points disappear with a bang. The point is that *all* saddle-node bifurcations have this character locally.

EXAMPLE 8.1.1:

The following system has been discussed by Griffith (1971) as a model for a genetic control system. The activity of a certain gene is assumed to be directly induced by two copies of the protein for which it codes. In other words, the gene is stimulated by its own product, potentially leading to an autocatalytic feedback process. In dimensionless form, the equations are

$$\begin{aligned}\dot{x} &= -ax + y \\ \dot{y} &= \frac{x^2}{1+x^2} - by\end{aligned}$$

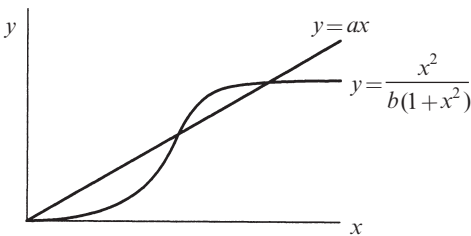
where x and y are proportional to the concentrations of the protein and the messenger RNA from which it is translated, respectively, and $a, b > 0$ are parameters that govern the rate of degradation of x and y .

Show that the system has three fixed points when $a < a_c$, where a_c is to be determined. Show that two of these fixed points coalesce in a saddle-node bifurcation when $a = a_c$. Then sketch the phase portrait for $a < a_c$, and give a biological interpretation.

Solution: The nullclines are given by the line $y = ax$ and the sigmoidal curve

$$y = \frac{x^2}{b(1+x^2)}$$

as sketched in Figure 8.1.3. Now suppose we vary a while holding b fixed. This is simple to visualize, since a is the slope of the line. For small a there are three



intersections, as in Figure 8.1.3. As a increases, the top two intersections approach each other and collide when the line intersects the curve tangentially. For larger values of a , those fixed points disappear, leaving the origin as the only fixed point.

Figure 8.1.3

To find a_c , we compute the fixed points directly and find where they coalesce. The nullclines intersect when

$$ax = \frac{x^2}{b(1+x^2)}.$$

One solution is $x^* = 0$, in which case $y^* = 0$. The other intersections satisfy the quadratic equation

$$ab(1+x^2) = x \quad (2)$$

which has two solutions

$$x^* = \frac{1 \pm \sqrt{1-4a^2b^2}}{2ab}$$

if $1 - 4a^2b^2 > 0$, i.e., $2ab < 1$. These solutions coalesce when $2ab = 1$. Hence

$$a_c = 1/2b.$$

For future reference, note that the fixed point $x^* = 1$ at the bifurcation.

The nullclines (Figure 8.1.4) provide a lot of information about the phase portrait for $a < a_c$. The vector field is vertical on the line $y = ax$ and horizontal on the sigmoidal curve. Other arrows can be sketched by noting the signs of $\dot{x}$ and $\dot{y}$. It appears that the middle fixed point is a saddle and the other two are sinks. To confirm this, we turn now to the classification of the fixed points.

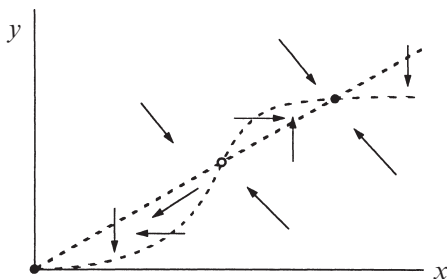


Figure 8.1.4

The Jacobian matrix at (x, y) is

$$A = \begin{pmatrix} -a & 1 \\ \frac{2x}{(1+x^2)^2} & -b \end{pmatrix}.$$

A has trace $\tau = -(a + b) < 0$ so all the fixed points are either sinks or saddles, depending on the value of the determinant Δ . At $(0,0)$, $\Delta = ab > 0$, so the origin is always a stable fixed point. In fact, it is a *stable node*, since $\tau^2 - 4\Delta = (a - b)^2 > 0$ (except in the degenerate case $a = b$, which we disregard). At the other two fixed points, Δ looks messy but it can be simplified using (2). We find

$$\Delta = ab - \frac{2x^*}{(1+(x^*)^2)^2} = ab \left[1 - \frac{2}{1+(x^*)^2} \right] = ab \left[\frac{(x^*)^2 - 1}{1+(x^*)^2} \right].$$

So $\Delta < 0$ for the “middle” fixed point, which has $0 < x^* < 1$; this is a *saddle point*. The fixed point with $x^* > 1$ is always a *stable node*, since $\Delta < ab$ and therefore $\tau^2 - 4\Delta > (a - b)^2 > 0$.

The phase portrait is plotted in Figure 8.1.5. By looking back at Figure 8.1.4, we can see that the unstable manifold of the saddle is necessarily trapped in the narrow channel between the two nullclines. More importantly, the *stable* manifold separates the plane into two regions, each a basin of attraction for a sink.

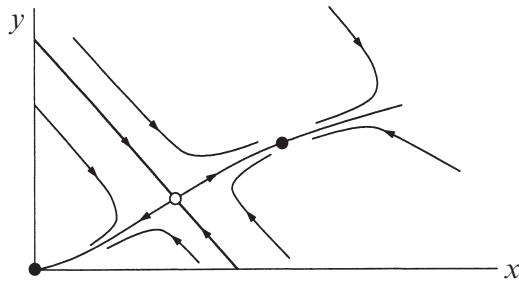


Figure 8.1.5

The biological interpretation is that the system can act like a *biochemical switch*, but only if the mRNA and protein degrade slowly enough—specifically, their decay rates must satisfy $ab < 1/2$. In this case, there are two stable steady states: one at the origin, meaning that the gene is silent and there is no protein around to turn it on; and one where x and y are large, meaning that the gene is active and sustained by the high level of protein. The stable manifold of the saddle acts like a threshold; it determines whether the gene turns on or off, depending on the initial values of x and y . ■

As advertised, the flow in Figure 8.1.5 is qualitatively similar to that in the idealized Figure 8.1.1. All trajectories relax rapidly onto the unstable manifold of the saddle, which plays a completely analogous role to the x -axis in Figure 8.1.1.

Thus, in many respects, the bifurcation is a fundamentally one-dimensional event, with the fixed points sliding toward each other along the unstable manifold

like beads on a string. *This is why we spent so much time looking at bifurcations in one-dimensional systems*—they’re the building blocks of analogous bifurcations in higher dimensions. (The fundamental role of one-dimensional systems can be justified rigorously by “center manifold theory”—see Wiggins (1990) for an introduction.)

Transcritical and Pitchfork Bifurcations

Using the same idea as above, we can also construct prototypical examples of transcritical and pitchfork bifurcations at a stable fixed point. In the x -direction the dynamics are given by the normal forms discussed in Chapter 3, and in the y -direction the motion is exponentially damped. This yields the following examples:

$$\begin{array}{lll} \dot{x} = \mu x - x^2, & \dot{y} = -y & \text{(transcritical)} \\ \dot{x} = \mu x - x^3, & \dot{y} = -y & \text{(supercritical pitchfork)} \\ \dot{x} = \mu x + x^3, & \dot{y} = -y & \text{(subcritical pitchfork)} \end{array}$$

The analysis in each case follows the same pattern, so we’ll discuss only the supercritical pitchfork, and leave the other two cases as exercises.

EXAMPLE 8.1.2:

Plot the phase portraits for the supercritical pitchfork system $\dot{x} = \mu x - x^3$, $\dot{y} = -y$, for $\mu < 0$, $\mu = 0$, and $\mu > 0$.

Solution: For $\mu < 0$, the only fixed point is a stable node at the origin. For $\mu = 0$, the origin is still stable, but now we have very slow (algebraic) decay along the x -direction instead of exponential decay; this is the phenomenon of “critical slowing down” discussed in Section 3.4 and Exercise 2.4.9. For $\mu > 0$, the origin loses stability and gives birth to two new stable fixed points symmetrically located at $(x^*, y^*) = (\pm\sqrt{\mu}, 0)$. By computing the Jacobian at each point, you can check that the origin is a saddle and the other two fixed points are stable nodes. The phase portraits are shown in Figure 8.1.6. ■

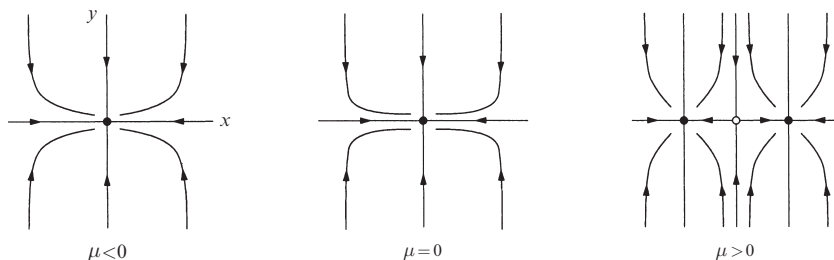


Figure 8.1.6

As mentioned in Chapter 3, pitchfork bifurcations are common in systems that have a symmetry. Here's an example.

EXAMPLE 8.1.3:

Show that a supercritical pitchfork bifurcation occurs at the origin in the system

$$\begin{aligned}\dot{x} &= \mu x + y + \sin x \\ \dot{y} &= x - y\end{aligned}$$

and determine the bifurcation value μ_c . Plot the phase portrait near the origin for μ slightly greater than μ_c .

Solution: The system is invariant under the change of variables $x \rightarrow -x, y \rightarrow -y$, so the phase portrait must be symmetric under reflection through the origin. The origin is a fixed point for all μ , and its Jacobian is

$$A = \begin{pmatrix} \mu + 1 & 1 \\ 1 & -1 \end{pmatrix}$$

which has $\tau = \mu$ and $\Delta = -(\mu + 2)$. Hence the origin is a stable fixed point if $\mu < -2$ and a saddle if $\mu > -2$. This suggests that a pitchfork bifurcation occurs at $\mu_c = -2$. To confirm this, we seek a symmetric pair of fixed points close to the origin for μ close to μ_c . (Note that at this stage we don't know whether the bifurcation is sub- or supercritical.) The fixed points satisfy $y = x$ and hence $(\mu + 1)x + \sin x = 0$. One solution is $x = 0$, but we've found that already. Now suppose x is small and nonzero, and expand the sine as a power series. Then

$$(\mu + 1)x + x - \frac{x^3}{3!} + O(x^5) = 0.$$

After dividing through by x and neglecting higher-order terms, we get $\mu + 2 - x^2/6 \approx 0$. Hence there is a pair of fixed points with $x^* \approx \pm\sqrt{6(\mu + 2)}$ for μ slightly greater than -2 . Thus a *supercritical* pitchfork bifurcation occurs at $\mu_c = -2$. (If the bifurcation had been subcritical, the pair of fixed points would exist when the origin was stable, not after it has become a saddle.) Because the bifurcation is supercritical, we know the new fixed points are stable *without even checking*.

To draw the phase portrait near $(0,0)$ for μ slightly greater than -2 , it's helpful to find the eigenvectors of the Jacobian at the origin. This can be done exactly, but a simple approximation is that the Jacobian is close to that at the bifurcation. Thus

$$A \approx \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

which has eigenvectors $(1,1)$ and $(1,-1)$, with eigenvalues $\lambda = 0$ and $\lambda = -2$, respectively. For μ slightly greater than -2 , the origin becomes a saddle and so the zero eigenvalue becomes slightly positive. This information implies the phase portrait shown in Figure 8.1.7.

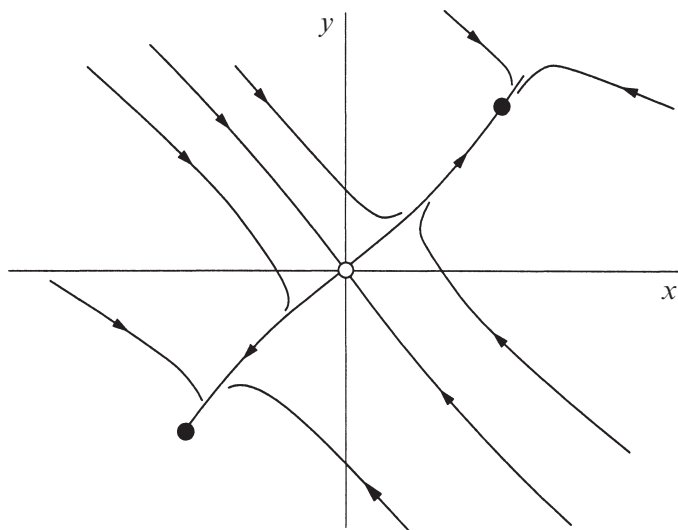


Figure 8.1.7

Note that because of the approximations we've made, this picture is only valid *locally* in both parameter and phase space—if we're not near the origin and if μ is not close to μ_c , all bets are off. ■

In all of the examples above, the bifurcation occurs when $\Delta = 0$, or equivalently, when one of the eigenvalues equals zero. More generally, the saddle-node, transcritical, and pitchfork bifurcations are all examples of **zero-eigenvalue bifurcations**. (There are other examples, but these are the most common.) Such bifurcations always involve the collision of two or more fixed points.

In the next section we'll consider a fundamentally new kind of bifurcation, one that has no counterpart in one-dimensional systems. It provides a way for a fixed point to lose stability without colliding with any other fixed points.

8.2 Hopf Bifurcations

Suppose a two-dimensional system has a stable fixed point. What are all the possible ways it could lose stability as a parameter μ varies? The eigenvalues of the Jacobian are the key. If the fixed point is stable, the eigenvalues λ_1, λ_2 must both lie in the left half-plane $\text{Re } \lambda < 0$. Since the λ 's satisfy a quadratic equation with

real coefficients, there are two possible pictures: either the eigenvalues are both real and negative (Figure 8.2.1a) or they are complex conjugates (Figure 8.2.1b). To destabilize the fixed point, we need one or both of the eigenvalues to cross into the right half-plane as μ varies.

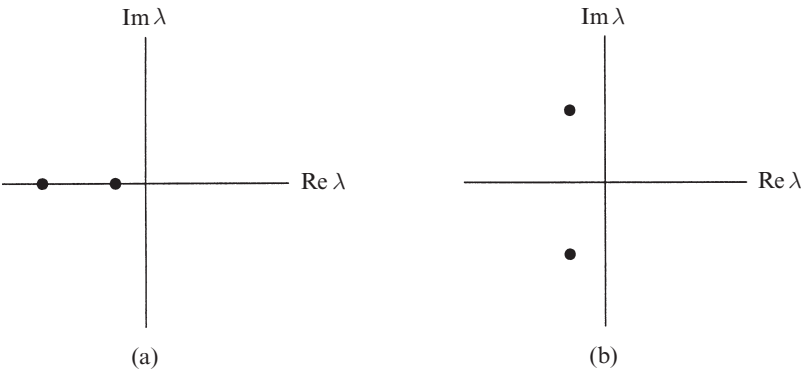


Figure 8.2.1

In Section 8.1 we explored the cases in which a real eigenvalue passes through $\lambda = 0$. These were just our old friends from Chapter 3, namely the saddle-node, transcritical, and pitchfork bifurcations. Now we consider the other possible scenario, in which two complex conjugate eigenvalues simultaneously cross the imaginary axis into the right half-plane.

Supercritical Hopf Bifurcation

Suppose we have a physical system that settles down to equilibrium through exponentially damped oscillations. In other words, small disturbances decay after “ringing” for a while (Figure 8.2.2a). Now suppose that the decay rate depends on a control parameter μ . If the decay becomes slower and slower and finally changes to *growth* at a critical value μ_c , the equilibrium state will lose stability. In many cases the resulting motion is a small-amplitude, sinusoidal, limit cycle oscillation about the former steady state (Figure 8.2.2b). Then we say that the system has undergone a *supercritical Hopf bifurcation*.

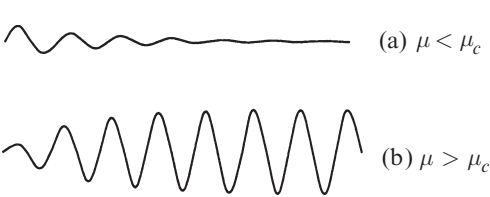


Figure 8.2.2

In terms of the flow in phase space, a supercritical Hopf bifurcation occurs when a stable spiral changes into an unstable spiral surrounded by a small, nearly elliptical limit cycle. Hopf bifurcations can occur in phase spaces

of any dimension $n \geq 2$, but as in the rest of this chapter, we'll restrict ourselves to two dimensions.

A simple example of a supercritical Hopf bifurcation is given by the following system:

$$\begin{aligned}\dot{r} &= \mu r - r^3 \\ \dot{\theta} &= \omega + br^2.\end{aligned}$$

There are three parameters: μ controls the stability of the fixed point at the origin, ω gives the frequency of infinitesimal oscillations, and b determines the dependence of frequency on amplitude for larger amplitude oscillations.

Figure 8.2.3 plots the phase portraits for μ above and below the bifurcation. When $\mu < 0$ the origin $r = 0$ is a stable spiral whose sense of rotation depends on the sign of ω . For $\mu = 0$ the origin is still a stable spiral, though a very weak one: the decay is only algebraically fast. (This case was shown in Figure 6.3.2. Recall that the linearization wrongly predicts a center at the origin.) Finally, for $\mu > 0$ there is an unstable spiral at the origin and a stable circular limit cycle at $r = \sqrt{\mu}$.

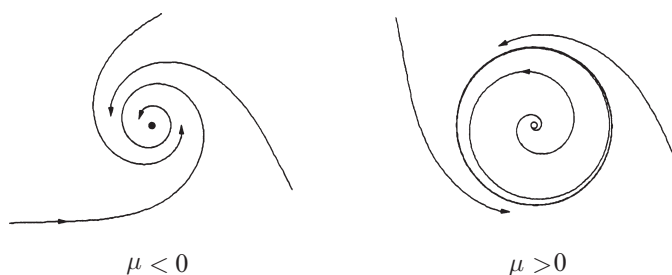


Figure 8.2.3

To see how the eigenvalues behave during the bifurcation, we rewrite the system in Cartesian coordinates; this makes it easier to find the Jacobian. We write $x = r \cos \theta$, $y = r \sin \theta$. Then

$$\begin{aligned}\dot{x} &= \dot{r} \cos \theta - r \dot{\theta} \sin \theta \\ &= (\mu r - r^3) \cos \theta - r(\omega + br^2) \sin \theta \\ &= (\mu - [x^2 + y^2])x - (\omega + b[x^2 + y^2])y \\ &= \mu x - \omega y + \text{cubic terms}\end{aligned}$$

and similarly

$$\dot{y} = \omega x + \mu y + \text{cubic terms}.$$

So the Jacobian at the origin is

$$A = \begin{pmatrix} \mu & -\omega \\ \omega & \mu \end{pmatrix},$$

which has eigenvalues

$$\lambda = \mu \pm i\omega.$$

As expected, the eigenvalues cross the imaginary axis from left to right as μ increases from negative to positive values.

Rules of Thumb

Our idealized case illustrates two rules that hold *generically* for supercritical Hopf bifurcations:

1. The size of the limit cycle grows continuously from zero, and increases proportional to $\sqrt{\mu - \mu_c}$, for μ close to μ_c .
2. The frequency of the limit cycle is given approximately by $\omega = \text{Im } \lambda$, evaluated at $\mu = \mu_c$. This formula is exact at the birth of the limit cycle, and correct within $O(\mu - \mu_c)$ for μ close to μ_c . The period is therefore $T = (2\pi/\text{Im } \lambda) + O(\mu - \mu_c)$.

But our idealized example also has some artifactual properties. First, in Hopf bifurcations encountered in practice, the limit cycle is elliptical, not circular, and its shape becomes distorted as μ moves away from the bifurcation point. Our example is only typical topologically, not geometrically. Second, in our idealized case the eigenvalues move on horizontal lines as μ varies, i.e., $\text{Im } \lambda$ is strictly independent of μ . Normally, the eigenvalues would follow a curvy path and cross the imaginary axis with nonzero slope (Figure 8.2.4).

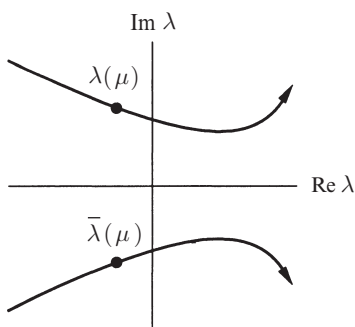


Figure 8.2.4

Subcritical Hopf Bifurcation

Like pitchfork bifurcations, Hopf bifurcations come in both super- and subcritical varieties. The subcritical case is always much more dramatic, and potentially dangerous in engineering applications. After the bifurcation, the trajectories must *jump* to a *distant* attractor, which may be a fixed point, another limit cycle, infinity, or—in three and higher dimensions—a chaotic attractor. We'll see a concrete example of this last, most interesting case when we study the Lorenz equations (Chapter 9).

But for now, consider the two-dimensional example

$$\begin{aligned}\dot{r} &= \mu r + r^3 - r^5 \\ \dot{\theta} &= \omega + br^2.\end{aligned}$$

The important difference from the earlier supercritical case is that the cubic term r^3 is now *destabilizing*; it helps to drive trajectories away from the origin.

The phase portraits are shown in Figure 8.2.5. For $\mu < 0$ there are two attractors, a stable limit cycle and a stable fixed point at the origin. Between them lies an unstable cycle, shown as a dashed curve in Figure 8.2.5; it's the player to watch in this scenario. As μ increases, the unstable cycle tightens like a noose around the fixed point. A **subcritical Hopf bifurcation** occurs at $\mu = 0$, where the unstable cycle shrinks to zero amplitude and engulfs the origin, rendering it unstable. For $\mu > 0$, the large-amplitude limit cycle is suddenly the only attractor in town. Solutions that used to remain near the origin are now forced to grow into large-amplitude oscillations.

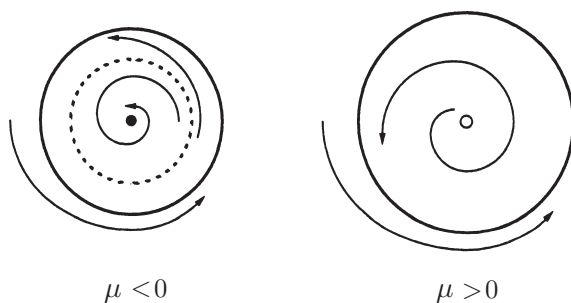


Figure 8.2.5

Note that the system exhibits *hysteresis*: once large-amplitude oscillations have begun, they cannot be turned off by bringing μ back to zero. In fact, the large oscillations will persist until $\mu = -1/4$ where the stable and unstable cycles collide and annihilate. This destruction of the large-amplitude cycle occurs via another type of bifurcation, to be discussed in Section 8.4.

Subcritical Hopf bifurcations occur in the dynamics of nerve cells (Rinzel and Ermentrout 1989), in aeroelastic flutter and other vibrations of airplane wings (Dowell and Ilgamova 1988, Thompson and Stewart 1986), and in instabilities of fluid flows (Drazin and Reid 1981).

Subcritical, Supercritical, or Degenerate Bifurcation?

Given that a Hopf bifurcation occurs, how can we tell if it's sub- or supercritical? The linearization doesn't provide a distinction: in both cases, a pair of eigenvalues moves from the left to the right half-plane.

An analytical criterion exists, but it can be difficult to use (see Exercises 8.2.12–15 for some tractable cases). A quick and dirty approach is to use the computer. If a small, attracting limit cycle appears immediately after the fixed point goes unstable, and if its amplitude shrinks back to zero as the parameter is reversed, the bifurcation is supercritical; otherwise, it's probably subcritical, in which case the nearest attractor might be far from the fixed point, and the system may exhibit hysteresis as the parameter is reversed. Of course, computer experiments are not proofs and you should check the numerics carefully before making any firm conclusions.

Finally, you should also be aware of a *degenerate Hopf bifurcation*. An example is given by the damped pendulum $\ddot{x} + \mu\dot{x} + \sin x = 0$. As we change the damping μ from positive to negative, the fixed point at the origin changes from a stable to an unstable spiral. However at $\mu = 0$ we do *not* have a true Hopf bifurcation because there are no limit cycles on either side of the bifurcation. Instead, at $\mu = 0$ we have a continuous band of closed orbits surrounding the origin. These are not limit cycles! (Recall that a limit cycle is an *isolated* closed orbit.)

This degenerate case typically arises when a nonconservative system suddenly becomes conservative at the bifurcation point. Then the fixed point becomes a nonlinear center, rather than the weak spiral required by a Hopf bifurcation. See Exercise 8.2.11 for another example.

EXAMPLE 8.2.1:

Consider the system $\dot{x} = \mu x - y + xy^2$, $\dot{y} = x + \mu y + y^3$. Show that a Hopf bifurcation occurs at the origin as μ varies. Is the bifurcation subcritical, supercritical, or degenerate?

Solution: The Jacobian at the origin is $A = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}$, which has $\tau = 2\mu$, $\Delta = \mu^2 + 1 > 0$, and $\lambda = \mu \pm i$. Hence, as μ increases through zero, the origin changes from a stable spiral to an unstable spiral. This suggests that some kind of Hopf bifurcation takes place at $\mu = 0$.

To decide whether the bifurcation is subcritical, supercritical, or degenerate, we use simple reasoning and numerical integration. If we transform the system to polar coordinates, we find that

$$\dot{r} = \mu r + r y^2,$$

as you should check. Hence $\dot{r} \geq \mu r$. This implies that for $\mu > 0$, $r(t)$ grows *at least* as fast as $r_0 e^{\mu t}$. In other words, all trajectories are repelled out to infinity! So there are certainly no closed orbits for $\mu > 0$. In particular, the unstable spiral is not surrounded by a stable limit cycle; hence the bifurcation cannot be supercritical.

Could the bifurcation be degenerate? That would require that the origin be a nonlinear center when $\mu = 0$. But $\dot{r}$ is strictly positive away from the x -axis, so closed orbits are still impossible.

By process of elimination, we expect that the bifurcation is *subcritical*. This is confirmed by Figure 8.2.6, which is a computer-generated phase portrait for $\mu = -0.2$.

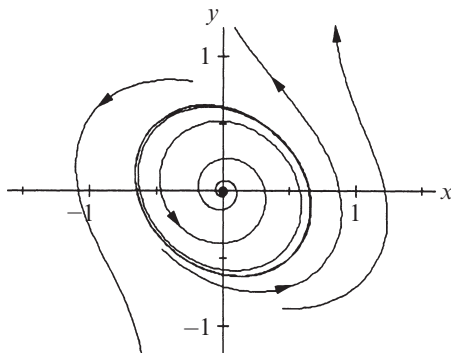


Figure 8.2.6

Note that an *unstable* limit cycle surrounds the stable fixed point, just as we expect in a subcritical bifurcation. Furthermore, the cycle is nearly elliptical and surrounds a gently winding spiral—these are typical features of *either* kind of Hopf bifurcation. ■

8.3 Oscillating Chemical Reactions

For an application of Hopf bifurcations, we now consider a class of experimental systems known as **chemical oscillators**. These systems are remarkable, both for their spectacular behavior and for the story behind their discovery. After presenting this background information, we analyze a simple model for oscillations in the chlorine dioxide–iodine–malonic acid reaction. The definitive reference

on chemical oscillations is the book edited by Field and Burger (1985). See also Epstein et al. (1983), Winfree (1987b) and Murray (2002).

Belousov's "Supposedly Discovered Discovery"

In the early 1950s the Russian biochemist Boris Belousov was trying to create a test tube caricature of the Krebs cycle, a metabolic process that occurs in living cells. When he mixed citric acid and bromate ions in a solution of sulfuric acid, and in the presence of a cerium catalyst, he observed to his astonishment that the mixture became yellow, then faded to colorless after about a minute, then returned to yellow a minute later, then became colorless again, and continued to oscillate dozens of times before finally reaching equilibrium after about an hour.

Today it comes as no surprise that chemical reactions can oscillate spontaneously—such reactions have become a standard demonstration in chemistry classes, and you may have seen one yourself. (For recipes, see Winfree (1980).) But in Belousov's day, his discovery was so radical that he couldn't get his work published. It was thought that all solutions of chemical reagents must go *monotonically* to equilibrium, because of the laws of thermodynamics. Belousov's paper was rejected by one journal after another. According to Winfree (1987b, p.161), one editor even added a snide remark about Belousov's "supposedly discovered discovery" to the rejection letter.

Belousov finally managed to publish a brief abstract in the obscure proceedings of a Russian medical meeting (Belousov 1959), although his colleagues weren't aware of it until years later. Nevertheless, word of his amazing reaction circulated among Moscow chemists in the late 1950s, and in 1961 a graduate student named Zhabotinsky was assigned by his adviser to look into it. Zhabotinsky confirmed that Belousov was right all along, and brought this work to light at an international conference in Prague in 1968, one of the few times that Western and Soviet scientists were allowed to meet. At that time there was a great deal of interest in biological and biochemical oscillations (Chance et al. 1973) and the BZ reaction, as it came to be called, was seen as a manageable model of those more complex systems.

The analogy to biology turned out to be surprisingly close: Zaikin and Zhabotinsky (1970) and Winfree (1972) observed beautiful propagating *waves* of oxidation in thin unstirred layers of BZ reagent, and found that these waves annihilate upon collision, just like waves of excitation in neural or cardiac tissue. The waves always take the shape of expanding concentric rings or spirals (Color plate 1). Spiral waves are now recognized to be a ubiquitous feature of chemical, biological, and physical excitable media; in particular, spiral waves and their three-dimensional analogs, "scroll waves", appear to be implicated in certain cardiac arrhythmias, a problem of great medical importance (Winfree 1987b).

Boris Belousov would be pleased to see what he started.